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EDUCATION

Jan 2023 – Aug 2026

University of British Columbia, Vancouver, Canada

Doctor of Philosophy in Forestry

Supervisor: Dr. Haibo Feng

Thesis title: Reshaping urban futures: geo-spatial strategies for urban net zero energy buildings

May 2018 – May 2020

University of British Columbia, Kelowna, Canada

Master of Science in Biology

Supervisor: Dr. Nathan Pelletier

Thesis title: Life cycle assessment of net zero energy poultry housing

Sept 2012 – Jun 2016

Beijing Normal University, Beijing, China

Bachelor of Environmental Engineering

Supervisor: Dr. Xitao Liu

RESEARCH INTERESTS

- Urban systems 2D/3D city modeling, urban building energy modeling (UBEM)
- Spatialized dynamic life cycle assessment (LCA)
- Application of geospatial technologies (GIS and remote sensing) in renewable urban buildings
- Net zero energy and emissions strategies for buildings, communities, and cities
- Urban green space/greening and district energy in building energy management

SUMMARY OF QUALIFICATIONS

- 7+ years' experience in LCA, EPD, and ESG in academic and industry settings
- 5+ years' interdisciplinary research on urban net zero energy buildings and geospatial modeling
- Consultant on GIS and building energy assessment for 3 Canadian municipalities; contributor to government sustainability reports
- Mentored 5 undergraduates and 2 master's students; teaching or TA experience in 5+ courses
- Led interdisciplinary teams and engaged with policymakers, industry, and communities
- Awarded 13 fellowships and scholarships during graduate studies
- Proficient in Energy Plus, City BES, ENVI-met, Revit, ArcGIS Pro, and Python, QGIS
- 3 international research visits; presented at 5 major conferences

SELECTED PUBLICATIONS

Li, Y., & Feng, H. (2025). Comprehensive spatial LCA framework for urban scale net zero energy buildings in Canada using GIS and BIM. *Applied Energy*, 388, 125649.

<https://doi.org/10.1016/j.apenergy.2025.125649>

Li, Y., & Feng, H. (2025). How geospatial technologies are transforming urban net-zero energy buildings: A comprehensive review of insights, challenges, and future directions. *Journal of Building Engineering*, 104, 112357. <https://doi.org/10.1016/j.job.2025.112357>

Li, Y., & Feng, H. (2025). Pathways to urban net zero energy buildings in Canada: A comprehensive GIS-based framework using open data. *Sustainable Cities and Society*, 122, 106263. <https://doi.org/10.1016/j.scs.2025.106263>

Li, Y., & Feng, H. (2025). Integrating urban building energy modeling (UBEM) and urban-building environmental impact assessment (UB-EIA) for sustainable urban development: A comprehensive review. *Renewable and Sustainable Energy Reviews*, 213, 115471.



<https://doi.org/10.1016/j.rser.2025.115471>

Li, Y., & Feng, H. (2023). GIS for the Potential Application of Renewable Energy in Buildings towards Net Zero: A Perspective. *Buildings*, 13(5), Article 5. <https://doi.org/10.3390/buildings13051205>

Li, Y., Arulnathan, V., Heidari, M. D., & Pelletier, N. (2022). Design considerations for net zero energy buildings for intensive, confined poultry production: A review of current insights, knowledge gaps, and future directions. *Renewable and Sustainable Energy Reviews*, 154, 111874.

Li, Y., Allacker, K., Feng, H., MD Heidari, & Pelletier, N. (2021). Net zero energy barns for industrial egg production: an effective sustainable intensification strategy? *Journal of Cleaner Production*(9), 128014.

Bamber, N., Turner, I., Arulnathan, V., Li, Y., & Pelletier, N. (2019). Comparing sources and analysis of uncertainty in consequential and attributional life cycle assessment: review of current practice and recommendations. *The International Journal of Life Cycle Assessment*.

PROFESSIONAL SERVICES

- Peer reviewer in Journal of Cleaner Production and Renewable and Sustainable Energy Reviews
- Collaborator with municipal and federal partners (Richmond, New Westminster, Kingston, NRC): Contributed to technical reports and policy recommendations, presenting research findings and integrating community and stakeholder's feedback
- Graduate Affiliate in Clean Energy Resource Centre, Bio-Products Institute, and BCCFA

RESEARCH EXPERIENCE

➤ 2023 – 2026 **Graduate Student/Research Assistant**

Sustainable Built Environment Lab, Canada

Research Title: Reshaping urban futures: geo-spatial strategies for urban net zero energy buildings

Funding source: Research supported by Dr. Feng's NSERC Discovery Grant

Supervisor: Dr. Haibo Feng. Committee Members: Dr. Keunhyun Park and Dr. Taraneh Sowlati

- Literature reviews on the applications of geospatial technologies in urban energy systems
- Developed GIS-based and urban building energy models
- Evaluated the impact of retrofit and renewable energy measures at urban building

➤ 2018-2020 **Graduate Student/Research Assistant**

Food Systems Priority Research for Integrated Sustainability Management Lab, Canada

Research title: Life cycle assessment of net zero energy poultry housing

Funding source: Research supported by Dr. Pelletier 's NSERC Discovery Grant and Pulse Canada Project

Supervisor: Dr. Nathan Pelletier. Committee Members: Dr. Kasun Hewage, Dr. Rehan Sadiq

- Undertook literature review about the application of net zero energy in Canadian agriculture
- Performed an ISO-14044-compliant, cradle-to-farm gate environmental life cycle assessment

➤ 2013 – 2016 **Research Assistant**

State Key Laboratory of Water Environment Simulation in Beijing Normal University, China

Funding source: Projects of National Natural Science Foundation

Supervisor: Dr. Xinghui Xia; Dr. Xitao Liu

Project 1: the bioaccumulation of PFASs in the water with dissolved organic matter of different forms and compositions (Funding #51279010)

Project 2: the degradation of Atrazine in the water redox systems that generate activated sodium persulfate (Funding #41671487)

- Literature review, designed experiments and conducted biochemical and chemical
- Experiments, Daphnia Magna and Scenedesmus Obliquus culture, instrumental analysis (HPLC, LC-MS, GC, and TOC detector), data analysis and model application (e.g., SPSS and Matlab)



PROJECTS

- 1) 2026 Benchmarking in Building Life Cycle Assessment, **National Research Council Canada**
- 2) 2025 Best Practices Research to Inform the Replacement and Design of Integrated Rooftop Systems, **City of New Westminster**, BC, Canada
- 3) 2024-2025 GIS-Social Mapping and Land Assessment for The Corporation of The City of Kingston Using HART Tool, **City of Kingston**, ON, Canada
- 4) 2024-2025 MATT Concrete “Straight Up” EPD project, **MATT Concrete**, BC, Canada
- 5) 2023 Developing A GIS-Based Urban Building Energy Assessment Model to Support Decision Making in Richmond, **City of Richmond**, BC, Canada
- 6) 2020 Spatially Resolved Inventory and Emissions Modelling for Pea and Lentil Life Cycle Assessment, **Pulse Canada** Project

FUNDING AND AWARDS

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| 2026 | Canada Postdoctoral Research Award (CPRA) |
| 2026 | Graduate Student Travel and Research Dissemination Fund |
| 2026 | Research-based Graduate Student Conference & Research Dissemination Fund |
| 2025-2026 | Dr. Imre Bella Graduate Scholarship in Forestry |
| 2025 | The High-Level Scientific Fellowship host by The Embassy of France in Canada |
| 2023-2026 | Faculty of Forestry Graduate Award |
| 2024-2025 | Donald S. McPhee Fellowship |
| 2024-2025 | WELWOOD of Canada Limited H. Richard Whittall Scholarship |
| 2024-2025 | Hamish Kimmins Scholarship in Forest Ecosystem |
| 2023 | VanDusen Graduate Fellowship in Forestry |
| 2023-2024 | International Tuition Award |
| 2023-2026 | the President's Academic Excellence Initiative PhD Award |
| 2023-2024 | Special UBC Graduate Scholarship-FRST Graduate Award |
| 2018-2020 | Graduate Research Associate Appointments |
| 2019 | University Graduate Fellowship |
| 2019 | The Mitacs Globalink Research Award (for a scholar visit in KU Leuven, Belgium) |
| 2018 | Graduate Deans Entrance Scholarship |

PRESENTATIONS

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| May 2026 | Oral Presentation, CSME–CFDSC–CSR, Vancouver, Canada |
| Nov 2025 | Oral Presentation, Smart and Sustainable Built Environment, Lille, France |
| Dec 2024 | Oral and poster presentations, Clean Energy Resource Centre, Vancouver, Canada |
| Aug 2024 | Oral Presentation, ISIE SEM/AP 2024 Conference, Beijing, China |
| May 2024 | Poster presentation, Bio-Products Institute, Vancouver, Canada |
| Oct 2023 | Poster presentation and competition (First-Prize Winner), CERC, Vancouver, Canada |
| Jun 2023 | Poster presentation, BCCFA, Kamloops, Canada |
| Mar 2020 | Oral presentation, Sustainability in Canadian Agriculture for Solutions, Saskatoon |
| Sept 2019 | Oral presentation and symposium presentation, LCA XIX ACLCA conference, Tucson, USA |
| Sept 2019 | Poster presentation, Biology Graduate Symposium, UBC, Canada |

SCHOLAR VISIT

- 1) Jun – Dec 2025 **Building Technology and Urban Systems Division, Lawrence Berkeley National Laboratory, USA**

Research topic: Green replacement/district energy system potentials in urban net zero energy

Supervisor: Dr. Tianzhen Hong



2) Nov – Dec 2025 **Civil Engineering (Génie Civil) Department, École Centrale de Lille, France**
Research topic: Developing a GIS-based model for assessing urban net-zero energy building
Funding source: The High-Level Scientific Fellowship host by The Embassy of France in Canada
Supervisor: Dr. Zoubair Lafhaj

3) May – Jul 2019 **Department of Architecture KU Leuven, Belgium**
Research topic: Life cycle inventory of net zero energy building
Funding source: The Mitacs Globalink Research Award
Supervisor: Dr. Karen Allacker

WORKING EXPERIENCE

- **Teaching Assistant** at the University of British Columbia – Vancouver & Kelowna, Canada
 - 2024-2026 FRST 490 / FRST 551 201 / LARC 542 001 Visual Resource Management and Planning for Aesthetics
 - 2023-2024 FRST 423 Forest Management Planning
 - 2023-2024 FRST 497 Communications and Professionalism
 - 2018-2019 Biology for Science Majors I
- **Guest Lecturer** at the University of British Columbia – Vancouver, Canada
 - 2025-2026 FRST 490 / FRST 551 201 / LARC 542 001 Visual Resource Management and Planning for Aesthetics
 - 2024 Bachelor of Science in Natural Resources - Bioeconomy Sciences & Technology (BEST) Program
 - 2025-2026 WOOD 491 Environmental Facilities Design
- **Associate** of ESG reporting at PwC, Guangzhou, China (2020)

SKILLS

Geospatial/urban: ArcGIS Pro, QGIS, remote sensing, LiDAR-based, 2D/3D city modelling

Building energy simulation: EnergyPlus, CityBES, OpenStudio, ENVI-met

BIM/design: Revit, SketchUp, AutoCAD

Environmental/data: OpenLCA, LCA, EPD, Python, MATLAB, SPSS, Java

Teaching development: UBC Course Design Intensive; Instructional Skills Workshop

REFERENCES

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| Dr. Haibo Feng | Department of Wood Science, Faculty of Forestry, University of British Columbia
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Email: haibo.feng@ubc.ca Phone: +1 604-822-9352 |
| Dr. Sheri Andrews-Key
(Reference for
teaching evaluation) | Director Micro-Certificate Programming CVA & CACE Micro-Certificate Lead, Faculty of Forestry, University of British Columbia
2424 Main Mall, Vancouver, BC Canada V6T 1Z4
Email: sheri.andrews-key@ubc.ca Phone: +1 306-801-1973 |
| Dr. Tianzhen Hong | Senior Scientist and Deputy Director for Research, Building and Industrial Energy Systems Division, Energy Technologies Area, Lawrence Berkeley National Lab
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